

ECON 301—ECONOMETRICS I

Spring 2020

Department of Economics
Nazarbayev University

Instructor: Alejandro Melo Ponce, 9.106, alejandro.melo@nu.edu.kz

Lecture time/Location: TR, 10:30 AM–11:45 AM, 8307

Office hours: By appointment, book at <https://alejandromelo.youcanbook.me>, please sign at least one day in advance.

TA: Assiya Nurzhanova, assiya.nurzhanova@nu.edu.kz

Course Description The purpose of this course is to help students understand how to interpret economic data. To that end, you will learn how to do basic empirical research. To do empirical research means using statistical techniques to manage and analyze data. The main goal is to quantify how one economic variable (such as someone's education) affects another variable (for instance his wage). These techniques are used by both academic economists (professors) and professional economists (i.e. those working for business, government and NGOs). Academic economists use them, in large part, to test theories and models; professional economists do empirical work to help management with decision making. You will gain practical experience by solving the problems sets where you will be analyzing actual economic data.

Course Requirements ECON 211—Economic Statistics, OR MATH 321—Probability and either MATH 322—Mathematical Statistics or MATH 310—Applied Statistical Methods.

If you have a working knowledge of the following terms, you are on good foot:

- Population, Sample, Data
- Random Variable, Probability Distribution
- Correlation, Independence
- Moments of a Distribution: Expected Value, Variance, Covariance, Correlation
- Point Estimator, Interval Estimator (Confidence Interval)
- Sampling Distribution of an Estimator
- Properties of Estimators: Unbiasedness, Consistency, Efficiency

Course Material The textbook for the class is *Introductory Econometrics: A Modern Approach* by Jeffrey Wooldridge, Cengage Learning, 2020, 7th Edition. I asked the library to provide the latest edition for you. However, any edition will suffice. There are several copies of older editions in the library.

Another recommended text is *Mastering 'Metrics, The path from cause to effect* by Joshua D. Angrist and Jorn-Steffen Pischke, Princeton University Press, 2014, 1st Edition. This is a very cheerful and enjoyable account of applied econometrics. It has a superb discussion of causal inference.

All other course materials and lecture notes will be posted in Moodle.

Computational Requirements R with RStudio are the recommended software packages. They are free to use. You are expected to familiarize yourself with this package without losing your focus on the main material in the course. If there is another software package you are already familiar with or more interested in learning (EViews, STATA, SAS, MATLAB, etc.), you might use that package for your homework assignments. Assignments as well as exams will include R output.

Course Grade There will be pop-up quizzes, homework assignments, a midterm exam, a final exam, and a research project.

There will be six pop-up quizzes. These quizzes will also serve as attendance checks. The minimum grade you can take from a quiz is 5 out of 10, which means that you will get 5 points even if your correct answers in the quiz are worth less than 5 points. If you arrive in the class five minutes later than the quiz begins, you will **not** be allowed to take it. The overall quiz grade will be calculated after dropping your lowest quiz grade.

There will be seven or eight homework assignments. The overall homework percentage will be calculated after dropping your lowest homework grade. Late submissions will **not** be accepted.

The midterm exam will take place between 7:00 PM–9:00 PM on **Thursday, February 20**. The final exam will be scheduled by the Registrar.

The final numeric grade will be calculated using the following weights:

Components	Weight
Pop-up Quizzes	10%
Homework Assignments	10%
Midterm	30%
Final Exam	35%
Research Project (more on this in the next section)	15%

The numeric grade will be curved to obtain the letter grade.

Students are expected to take all exams at the designated time and place. Make-up exams and quizzes will **not** be given. If you miss a midterm exam (or a quiz) due to serious illness, or injury of yourself or a close family member, its weight will be shifted to the remaining exams (quizzes) equally.

Research Project Finding a topic is an important part of the exercise: I won't offer suggestions (although I will be happy to respond to preliminary ideas that you come up with). The paper should be short: 5–10 pages (excluding title and reference pages) discussing the econometric analysis of a group of economic or socioeconomic variables of your choice. It should be well written. You are encouraged to use the Writing Center for help with this.

You are encouraged to work together for your research project **with a limit of maximum 3 people** per project. The project is due by **Friday, April 17, at 5 PM**. The research project grade is determined using the following rubric:

Components	Weight
Introduction and motives for the research	15%
Integration of the literature review into the paper	15%
Model and pretests	20%
Results and inference	20%
Conclusions	15%
Grammar, writing quality	15%

Course Outline

1. Introduction: What is Econometrics?
 - Chapter 1 & Appendix A
2. Simple Regression Model
 - Chapter 2 & Appendix B
3. Multiple Regression Analysis: Estimation

- Chapters 3
- 4. Inference
 - Chapters 4
- 5. OLS Asymptotics
 - Chapter 5
- 6. Further Issues
 - Chapter 6
- 7. Binary (or Dummy) Variables
 - Chapter 7
- 8. Heteroskedasticity
 - Chapter 8
- 9. More on Specification and Issues
 - Chapter 9

Rules and Regulations During lectures, please keep your laptops, tablets and cell phones switched off and refrain from text messaging. Do not arrive late or leave early without prior permission. Do not engage in side conversations or other disruptive behaviors.

Academic Misconduct Plagiarism or cheating will not be tolerated and sanctioned according to the procedures in the Student Code of Conduct for Nazarbayev University. Students should familiarize themselves with this code. Please note that I do not discuss plagiarism cases unless a student makes an appeal to the disciplinary committee where my original penalty might be decreased or increased.

You can find the university's official statement on plagiarism below:

Plagiarism is intentionally or carelessly presenting the work of another as one's own. It includes submitting an assignment purporting to be the student's original work which has wholly or in part been created by another person. It also includes the presentation of the work, ideas, representations, or words of another person without customary and proper acknowledgement of sources. Plagiarism occurs when a person:

- 1. Directly copies one or more sentences of another person's written work without proper citation. If another writer's words are used, you must place quotation marks around the quoted material and include a footnote or other indication of the source of the quotation. This includes cut and paste from the Internet or other electronic sources;*
- 2. Changes words but copies the sentence structure of a source without giving credit to the original source, or closely paraphrases one or more paragraphs without acknowledgement of the source of the ideas, or uses graphs, figures, drawings, charts or other visual/audio materials without acknowledging the source or the permission of the author;*
- 3. Submits false or altered information in any academic exercise. This may include making up data for an experiment, altering data, citing nonexistent articles, contriving sources, etc.;*
- 4. Turns in all or part of assignment done by another student and claims it as their own;*
- 5. Uses a paper writing service, has another student write a paper, or uses a foreign language translation and submits it as their own original work.*